ELIAS COLESHILL

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PROFESSIONAL SUMMARY

Third-year Mechatronics and Artificial Intelligence Systems Engineering student with experience in robotic welding, PLC programming, embedded systems, and CAD. Completed a production engineering co-op at FIO Automotive Canada and lead a 35-member university robotics club.

EDUCATION

Bachelor of Engineering Science Expected Dec 2029

Western University | London, Ontario

Mechatronics and Artificial Intelligence Systems Engineering

- President of WE FIRST Robotics (2025-Present)
- Dean's List (2024, 2025)
- Member of Sunstangs Solar Car Team, WE FIRST Robotics, Western Engineering Competition, and the Western Climbing Club (2024-Present)

TECHNICAL SKILLS

Programming: Python, C++, Java, 6502 assembly

Embedded Hardware: Arduino, ESP32 / ESP32-C3, Jetson Orin Nano, RoboRIO, MOS6502

Industrial Controls: Mitsubishi PLC programming, Yaskawa 6-axis robotic arm teaching

Development and Analysis: GitHub, VS Code, MATLAB, Simulink, Excel

CAD & PCB Design: SolidWorks (CSWA), Onshape, AutoCAD, Altium, KiCad

3D Printing: Bambu Printers, Prusa Printers & Slicer, Cura Slicer, Formlabs SLA & PreForm

Design and Fabrication: DFM, GD&T, R&D, prototyping, root cause analysis, milling, turning, TIG, woodworking

WORK EXPERIENCE

Production Engineering Co-op May 2026-Aug 2026

FIO Automotive Canada | Stratford, ON

- Designed and implemented mechanical and controls upgrades for multiple six-axis robotic welding cells producing Toyota and Lexus sheet-metal components.
- Diagnosed production-quality issues using root-cause analysis, then implemented equipment redesigns and control-program changes.
- Supported Yaskawa and Panasonic welding robot teaching and Mitsubishi PLC programming in weld cells.
- Produced SolidWorks designs and coordinated equipment upgrades from concept and fabrication through testing and installation.
- Developed sensor-equipped chutes for finished-part checks, reducing cycle time from 61 sec to 55 sec.
- Collaborated with production, maintenance, quality, engineering, and external contractors to complete upgrades in an active manufacturing environment.
- Experience working on large engineering teams and working on cross functional teams to complete projects.

Robotics Teacher Oct 2022-Present

Self-Employed | Barrie, ON

Bike Mechanic Apr 2025-Aug 2025

The Bike Zone | Barrie, ON

Maintenance Crew Member Apr 2021-Jun 2024

Red Barn Event Center | Barrie, ON

ENGINEERING PROJECTS & LEADERSHIP

Humanoid Robot Summer 2026

- Lead a 12-member team developing the upper body of an AI-enabled humanoid robot.
- Designed the robot's distributed architecture using a Jetson Orin Nano for AI processing and ESP32 controllers for real-time motion control.
- Integrated LLM API calls with conversational body-language movements, allowing the robot to respond verbally and physically to users.
- Developed custom PCBs, power systems, 3D-printed components, and Onshape assemblies for the robot's mechanical and electrical subsystems. *Technologies:* Jetson Orin Nano, ESP32, Custom PCBs, 3D Printed Components, CAD Design in Onshape

Weather Activated Lights Spring 2026

- Designed and built an ESP32-based lighting system that retrieves forecast data through a weather API and communicates upcoming conditions using colour and animation patterns.
- Programmed a non-blocking C++ control system with custom temperature-to-colour mapping, cloud-based brightness adjustment, timed transitions, and deep-sleep scheduling.
- Designed and assembled a custom PCB integrating the ESP32, power circuitry, and LED controls within a 3D-printed enclosure.
- Developed schedule-aware operation that automatically activates the display before the user's morning routine.

President — WE FIRST Robotics Club 2025-Present

- Lead an eight-member executive team and a university robotics club of more than 35 members.
- Direct technical projects including an FRC team, 3 lb combat robots, workshops, and engineering competitions.
- Organize the annual Robot in 3 Days event for more than 25 participants, coordinating facilities, materials, schedules, and technical support.
- Manage project selection, team assignments, safety expectations, competition preparation, and club operations.

ACCOMPLISHMENTS

- WEC Senior Design - 2nd Place (2025)
- WEC Junior Design – 3rd Place (2024)
- FIRST Robotics Team 4617 Lead Mentor (2025-Present)

CERTIFICATIONS

CSWA | Onshape Parts & Assemblies | Microsoft Office | CPR/First Aid

ICT SHSM (includes Project Management, Basic Electrical Safety, Portfolio Development, etc.)